

B.E. I Yr. (Common to E&TC A, E&TC B)

Mid Test-III

Sub: Electrical Engineering (2EIRC4)

Time: 70 min

Max. Marks: 20

1. Explain the two-wattmeter method in detail for a balanced 3 phase star or delta connected load along with its necessary circuit, derivation and phasor diagrams. In a balance three phase 400 V circuit, the line current is 115.5 A when the power is measured by two wattmeter method, one meter reads 40 KW and other reads zero. What is the power factor of the load? If the power factor were unity and the line current is same, what would be the reading of each wattmeter. 3+3
2. How Armature reaction and commutation affects the performance of a dc machine. 4
3. Write short note on Speed control of a DC machine. 5
4. Discuss about different characteristics of a DC generator. 5